

EXPERIMENT 9

IMPLEMENTATION OF 2DES

AIM

To implement a simple Double DES (2DES) encryption and decryption using two keys in C.

ALGORITHM

1. Start the program.
2. Read the plaintext.
3. Read two secret keys.
4. Encrypt the plaintext using Key 1.
5. Encrypt the result again using Key 2.
6. Decrypt by applying Key 2 and then Key 1.

CODE

```
#include <stdio.h>
#include <string.h>

int main()
{
    char text[100], k1, k2;
    unsigned char enc[100];
    int i, len;

    printf("Enter Plain Text: ");
    fgets(text, sizeof(text), stdin);
    text[strcspn(text, "\n")] = '\0';

    printf("Enter Key 1: ");
    scanf(" %c", &k1);

    printf("Enter Key 2: ");
    scanf(" %c", &k2);

    len = strlen(text);

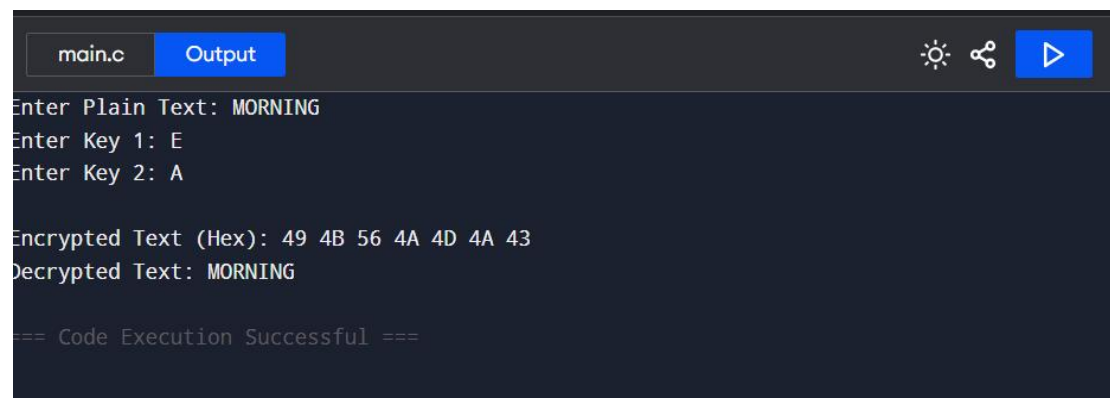
    printf("\nEncrypted Text (Hex): ");
    for(i=0;i<len;i++)
    {
```

```
        enc[i] = (text[i] ^ k1) ^ k2;
        printf("%02X ", enc[i]);
    }

    printf("\nDecrypted Text: ");
    for(i=0;i<len;i++)
    {
        printf("%c", (enc[i] ^ k2) ^ k1);
    }

    return 0;
}
```

OUTPUT



```
main.c Output
Enter Plain Text: MORNING
Enter Key 1: E
Enter Key 2: A

Encrypted Text (Hex): 49 4B 56 4A 4D 4A 43
Decrypted Text: MORNING

=== Code Execution Successful ===
```

RESULT

The plaintext was successfully encrypted and decrypted using a simple Double DES (2DES) simulation.